

02LQD Specification and Simulation of Digital Systems

2019/09/20

1. Combinational circuit design (4 + 2 + 2 points)

A car has a fuel level detector that outputs the current fuel level as a 3-bit binary number *lev*, where 000 means empty and 111 means full. Design a combinational circuit that illuminates a “low fuel” indicator light by setting output *L* to 1 when the fuel level drops below 3 (truth table, Karnaugh map, minimized Boolean expression) (4 points). Describe the circuit in VHDL according to the following description styles:

- dataflow using conditional signal assignments (WHEN ELSE statements) (2 points)
- dataflow using selected signal assignments (WITH SELECT WHEN statements) (2 points)

2. FSM-D design (22 points)

Design an alarm system that sets a single output bit *alarm* to 1 at each clock tick when the average temperature of 4 consecutive samples exceeds a user-defined threshold value, otherwise it is set to 0. A 16 bit unsigned input *CT* indicates the current temperature and a 16-bit unsigned input *WT* indicates the warning threshold. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (10 points)
2. Design a FSM-D (12 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Monday September 23 7pm**
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
 - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.